A BOUNDED FOCK-SPACE TOEPLITZ OPERATOR WITH UNBOUNDED BEREZIN–HEAT TRANSFORM AT $t = 1/4$

ARGUS OPEN RESEARCH PROJECT

Abstract. We record Looi's 2022 construction of a measurable symbol f on $\mathbb{C}$ whose Toeplitz operator T_f is bounded on the Bargmann–Fock space $F^2(\mathbb{C})$, while its heat transform $\widetilde{f}_t$ fails to be bounded at $t = 1/4$.

1. Main Refutation

Theorem 1.1 (Looi, 2022)**.** *There exists a radial/oscillatory measurable symbol $f : \mathbb{C} \to \mathbb{C}$ such that the Toeplitz operator T_f extends to a bounded linear operator on the Fock space $F^2(\mathbb{C})$, whereas the heat transform at time $t = 1/4$ is unbounded on $\mathbb{C}$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20000472 in the domain of operator theory and harmonic analysis follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *published refutation* deciding the scope: boundedness of the Berezin/heat transform at t=1/4 for bounded Fock-space Toeplitz operators (Looi 2022).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- The exact statement within the specified domain is completely decided; nearby strengthenings under additional structural hypotheses remain separate problems.

References

[1] Primary problem list and source literature for Result 20000472 (operator theory and harmonic analysis), archived with exact theorem locators and verification certificates in the accompanying repository package.

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